Parimala Pasupuleti

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PROFESSIONAL SUMMARY

Data and AI engineer with 8+ years shipping production systems at scale - **SQL optimization across 200+ queries (30% performance gain)**, ETL pipelines on **1.6M financial records**, AWS infrastructure (S3, EC2, RDS, Glue), and automated reporting pipelines eliminating 80% of manual operational workload. Built production **AI-augmented analytics systems** - Leasel (8-agent AI platform with reproducible eval harness, 93.6% precision) and AI on FHIR (NLP-driven clinical data retrieval, ~60% reduction in manual retrieval time). Strong background in **data governance, quality controls, and compliance-grade pipeline delivery** across regulated financial datasets. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

CORE COMPETENCIES

SQL & ETL Pipelines | AWS Data Infrastructure | Automated Reporting | Data Governance & QA | BI Dashboards & Visualization | Large-Scale Dataset Processing | AI-Augmented Analytics | Root-Cause Analysis

WORK EXPERIENCE

Entrata - PropTech / Real Estate SaaS

Sep 2018 - Aug 2024

Senior Software Engineer - Data & Analytics (Remote)

- **Automated 80% of manual reporting workload** via Python and PL/SQL pipelines - extraction, transformation, and delivery across 20+ production modules for 100K+ MAU product; 120+ Agile sprints across 6 years.
- **Optimized 200+ SQL queries** (30% performance gain across production reporting stack); root-cause analysis across complex multi-schema PostgreSQL, MySQL, and Oracle databases; 99.5% uptime over 6+ consecutive months.
- **Built and owned ETL pipelines on 1.6M financial records** (HUD Tax Credit, LIHTC, AMI-based pricing, HCV) - output validation gates, data quality controls, and compliance guardrails at launch; 100% PHI compliance on AWS S3 migration, zero security incidents.
- **AWS data infrastructure:** S3, EC2, RDS, and Glue for data pipeline and reporting architecture across 20+ modules; Docker microservices redesign (50% scalability improvement, 30% deployment time reduction).
- Mentored 5+ junior engineers across 300+ peer code reviews; defined data standards and KPI definitions for cross-functional stakeholder reporting across compliance, finance, and operations teams.

George Mason University - Research

Aug 2024 - Aug 2025

Research Associate - Data Analytics

- Built **multi-source prediction models across 190+ countries** (Python, R) on WHO and UK Road Safety datasets - R^2=0.977; applied Isolation Forest anomaly detection surfacing issues manual review missed across 3 research pipelines.
- Designed **Tableau, ggplot2, and Quarto dashboards** for faculty leadership - 40% research comprehension improvement across two cohorts; translated complex multi-dimensional datasets into stakeholder-ready visualizations.

Magizhs Tech Solution

May 2025 - Aug 2025

AI/ML Engineer Intern

- Built an **AI-augmented agentic analytics system** using Python and LangGraph - RAG layer over regulatory corpus; confidence-scored output with validation guardrails; leveraged AI-powered development tools throughout the build cycle.
- Integrated GenAI and LLM inference APIs into data retrieval and validation workflows - demonstrating the AI-augmented development approach the WWPS RACE team is building toward.

George Mason University - College of Engineering

Aug 2025 - May 2026

Graduate Teaching Assistant (Part-time)

- Delivered technical instruction to 600+ students across Python programming, database design, and SQL - weekly lab instruction, one-on-one debugging support, and rubric-aligned grading across multiple sections.

Entrata India - Pune

Jan 2017 - Sep 2018

Associate Software Engineer / Trainee

- Built ETL pipelines for 1.6M HUD/Tax Credit records; full-stack property management modules (PHP, Python, PostgreSQL); cloud migration to AWS S3; redesigned document storage database architecture.

PROJECTS

Leasel: AI-Augmented Data Intelligence Platform 2026

Built an 8-agent AI platform delivering **automated data intelligence and root-cause analytics** over regulated document corpora: RAG-based retrieval (91% Precision@1), reproducible evaluation harness (225-query benchmark, LLM-as-judge scoring, regression gating), and confidence-scored outputs with hard data quality boundaries. Demonstrates AI-augmented analytics: LLM handles reasoning and hypothesis generation; deterministic logic enforces every final data-backed conclusion. Production precision: 93.6%.

Python · LangGraph · FastAPI · RAG · SQLite · LLM-as-Judge eval harness · AI-augmented analytics

AI on FHIR: NLP-Driven Medical Records Query System 2024

Built a Flask + React NLP platform translating natural language queries (EN/ES) to FHIR APIs using LLMs, BERT, and embeddings - reduced manual clinical data retrieval time ~60% and enforced HIPAA-compliant RBAC across structured medical records.

Python · Flask · LLMs · FHIR · React · spaCy · BERT · Embeddings · Docker

EDUCATION

M.S. Data Analytics Engineering · George Mason University, Virginia May 2026

GPA: 3.87

B.Tech Computer Science & Engineering · Jawaharlal Nehru Technological University 2017

SKILLS

Languages: Python, SQL, PL/SQL, R, JavaScript, TypeScript, PHP, Bash

Data & BI: SQL (PostgreSQL, MySQL, Oracle, Redshift-adjacent), ETL Pipelines, AWS Glue, Data Modeling, Automated Reporting, Root-Cause Analysis, Data Governance, Data Quality Controls, Amazon QuickSight (learning), Tableau, Power BI, ggplot2, Quarto, Pandas, NumPy, Scikit-learn, XGBoost, Isolation Forest, Feature Engineering

AI & Analytics: AI-Augmented Development, GenAI Integration, LLM Inference APIs, LangGraph, RAG Pipelines, LLM Evaluation, LLM-as-Judge, Embeddings, OpenAI API, FastAPI, ChromaDB, spaCy, BERT

Cloud & Infrastructure: AWS (S3, EC2, RDS, Glue, DynamoDB, NoSQL), Azure, Docker, CI/CD (GitHub Actions), Microservices, RabbitMQ, Git, Linux, Agile/Scrum